DATA ANALYTICS PROJECT DOCUMENT

Title : Space Debris

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Organization : Anudip Foundation

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ABSTRACT

The rapid growth of satellite launches has led to an increase in space debris, raising concerns about orbital congestion and collision risks. This project analyzes space debris data to understand distribution patterns, congestion risk, and orbital behavior across different regions.Data from multiple sources was collected, cleaned, and integrated to perform exploratory analysis, visualization, and statistical testing. The results show that while altitude has minimal impact, orbital bands provide more meaningful insights into congestion risk.The study highlights the importance of data-driven analysis in identifying high-risk zones and supports better decision-making for safe and sustainable space operations.

INTRODUCTION

The project addresses the growing problem of space debris accumulation, which poses significant threats to satellites, space missions, and overall space sustainability.

With the rapid increase in satellite launches, the amount of debris in Earth's orbit has risen substantially, leading to higher risks of collisions and orbital congestion. This study focuses on analyzing the distribution and characteristics of space debris to better understand congestion risk across different orbital regions.

It involves a comprehensive analysis of key parameters such as altitude, orbital band, congestion risk, and inclination. Through this, the project identifies patterns in debris distribution and risk levels within various orbits.

Additionally, visualization techniques and statistical methods, including correlation, regression, and hypothesis testing, are applied to derive meaningful insights and support data-driven conclusions.

PROBLEM STATEMENT

The rapid expansion of space activities, driven by increasing satellite launches for communication, navigation, and observation, has led to a significant rise in space debris orbiting the Earth. This accumulation of debris has created highly congested orbital regions, increasing the risk of collisions between active satellites and non-functional objects. Such collisions can result in severe consequences, including damage to critical space infrastructure, disruption of services like GPS and communication systems, and the generation of even more debris, further worsening the problem.

Despite the availability of tracking systems that monitor space objects, there is a lack of effective analytical approaches to understand debris distribution, congestion patterns, and associated risk levels across different orbital regions. Existing solutions primarily focus on tracking positions rather than providing insights into how debris is clustered or how risk varies with factors such as altitude, orbital band, and inclination.

This project aims to address this gap by analyzing space debris data to identify patterns in congestion and risk distribution. By applying data preprocessing, visualization, and statistical techniques, the study seeks to provide meaningful insights that can support better decision-making for satellite operations and contribute to safer and more sustainable use of space.

OBJECTIVES

- To analyze space debris data to understand congestion patterns
- To study how risk varies across different orbital regions
- To visualize data for clear insights
- To apply statistical methods to validate findings
- To identify high-risk zones for safer space operations

BUSINESS CONTEXT

1. **Real-World Impact:** Helps in understanding orbital congestion and supports decisions to reduce collision risks and improve satellite safety

2. **Industry Domain:** Space Technology / Aerospace / Environment

3. **Stakeholders:** Space agencies (NASA, ISRO), satellite operators, private space companies

4. **Scale:** Large-scale dataset of thousands of space objects across multiple orbital regions

5. **Business Question:**
   "Which orbital bands are most crowded, and how can this impact future satellite deployments?"

6. **Potential Outcome / ROI:**
   Improved mission planning, reduced risk of satellite damage, and better long-term sustainability of space operations

METHODOLOGY

1. Data Collection

Space debris data was collected from multiple reliable sources, including datasets containing orbital parameters, object characteristics, and risk classifications. These datasets provided information such as altitude, inclination, orbital band, and congestion risk, forming the foundation for analysis.

2. Data Preprocessing

The collected data was cleaned to ensure quality and consistency. This involved handling missing values by removing incomplete records, eliminating duplicate entries, and standardizing formats such as column names and categorical values. Data types were also corrected to enable accurate analysis.

3. Data Integration

Multiple datasets were combined using a common key (object_id) to create a unified dataset. This integration allowed linking orbital parameters with object details and risk information, enabling a more comprehensive analysis.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and distribution of the data. Key variables such as altitude, orbital band, inclination, and congestion risk were analyzed to identify patterns, trends, and anomalies across different orbital regions.

5. Visualization

Various visualization techniques were used to represent insights clearly. Pie charts and donut charts were used for categorical distributions, dot plots for comparison of averages, and other appropriate graphs to highlight patterns and relationships in the data.

6. Statistical Analysis

Statistical methods were applied to validate findings. Correlation analysis was used to study relationships between variables, regression analysis to assess the impact of altitude on congestion risk, and hypothesis testing to determine statistical significance of observed patterns.

7. Result Interpretation

The final step involved interpreting the analytical results to derive meaningful insights. Key findings related to congestion patterns, risk distribution, and orbital behavior were summarized to support better understanding and decision-making for space debris management.

TOOLS & TECHNOLOGIES

1. **Programming Language :** Python in (Jupyter Notebook)

2. **Libraries Used** :
   - Pandas → Data manipulation and preprocessing
   - NumPy → Numerical computations
   - Matplotlib → Data visualization
   - Seaborn → Advanced visualization

3. **Data Handling Formats :** CSV datasets

4. **Dashboards** : Power BI

5. **Queries for Analysis** : MySql workbench.

DATA SOURCES

1. Satellite Orbital Data

Source: CelesTrak / Space-Track
Dataset: Satellite TLE (Two-Line Element) Data
Granularity: Object-level, real-time / periodic updates
Key Features : Mean Motion , Eccentricity , Inclination , Epoch , Object ID.

2. Satellite Characteristics Data

Source: Union of Concerned Scientists / Kaggle
Dataset: Satellite Database
Granularity: Satellite-level
Key Features : Country ,Purpose ,Orbital , Launch details.

3. Space Environment & Orbit Data

Source: NASA Open Data Portal
Dataset: Orbital Debris / Space Object Data
Granularity: Object-level
Key Features : Altitude ,Orbital Band , Congestion Indicators.

DATA CLEANING & PREPROCESSING

1. Handling Missing Values

The datasets were examined for missing or null values in critical columns such as altitude, inclination, and congestion risk. Rows with essential missing information were removed to maintain data integrity, while minor gaps were handled by excluding them from specific analyses to avoid distortion of results.

2. Removing Duplicate Records

Duplicate entries based on object_id were identified and removed to ensure that each space object was represented only once. This prevented bias and overcounting in analysis and visualizations.

3. Standardizing Column Names and Formats

Column names were cleaned by removing extra spaces and converting them to a consistent format (e.g., lowercase). Inconsistent naming such as OBJECT_ID and object_id was standardized to enable smooth merging and processing.

4. Data Type Conversion

Data types were corrected for accurate analysis. For example:

- object_id was converted to string for merging
- Numerical columns like altitude_km and inclination were converted to numeric types

This ensured compatibility during calculations and statistical analysis.

5. Cleaning Categorical Values

Categorical columns like congestion_risk and orbital_band were cleaned by:

- Removing extra spaces
- Standardizing text format (“high”, “ High ” → “High”)

This avoided duplicate categories during grouping.

6. Handling Outliers

Extreme or unrealistic values in numerical columns (such as very high or negative altitude) were checked. Outliers were either removed or retained based on their relevance to real-world scenarios, ensuring they did not skew the analysis.

7. Ensuring Data Consistency

Consistency checks were performed across datasets to ensure values aligned logically (e.g., altitude ranges matching orbital bands). Any inconsistencies were corrected or filtered out.

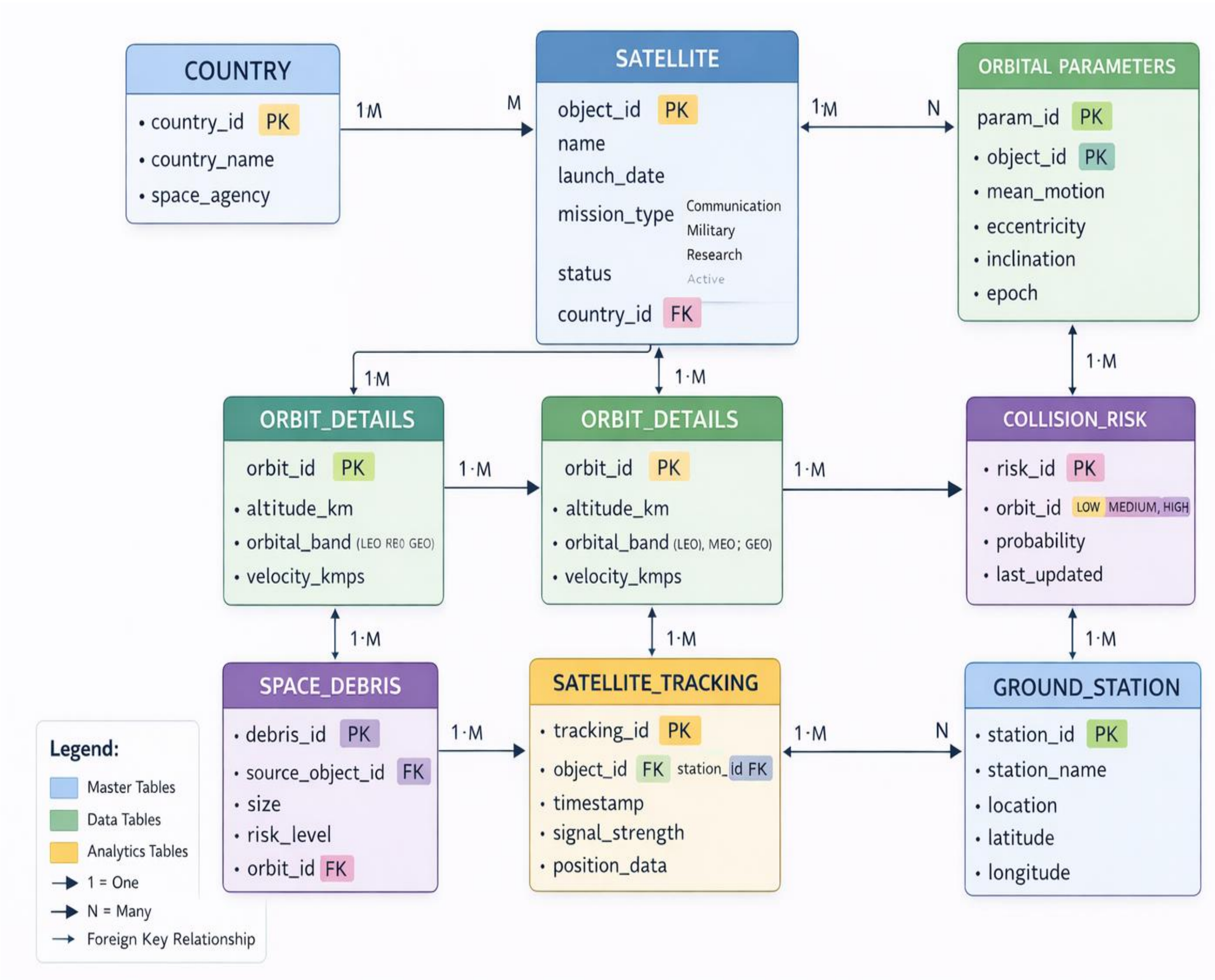
8. Data Integration Preparation

Before merging datasets, key columns (object_id) were standardized in format and type to ensure successful joins without data loss.

9. Final Clean Dataset Creation

After cleaning, a final structured dataset was created containing relevant features such as altitude, orbital band, inclination, and congestion risk, ready for analysis and visualization.

EXPLORATORY DATA ANALYSIS



Relationships Summary

- COUNTRY → SATELLITE = One country has many satellites.
- SATELLITE → ORBITAL_PARAMETERS = One satellite has many orbital records.
- ORBIT_DETAILS → SPACE_DEBRIS = One orbit may contain many debris objects.
- ORBIT_DETAILS → COLLISION_RISK = One orbit can have many collision risk records.
- GROUND_STATION → SATELLITE_TRACKING = One station tracks many satellites.
- SATELLITE → SATELLITE_TRACKING = One satellite can have many tracking logs.

Orbital Bands Used

- LEO → Low Earth Orbit
- MEO → Medium Earth Orbit
- GEO → Geostationary Earth Orbit

Purpose of This ER Diagram

This database system is designed to:

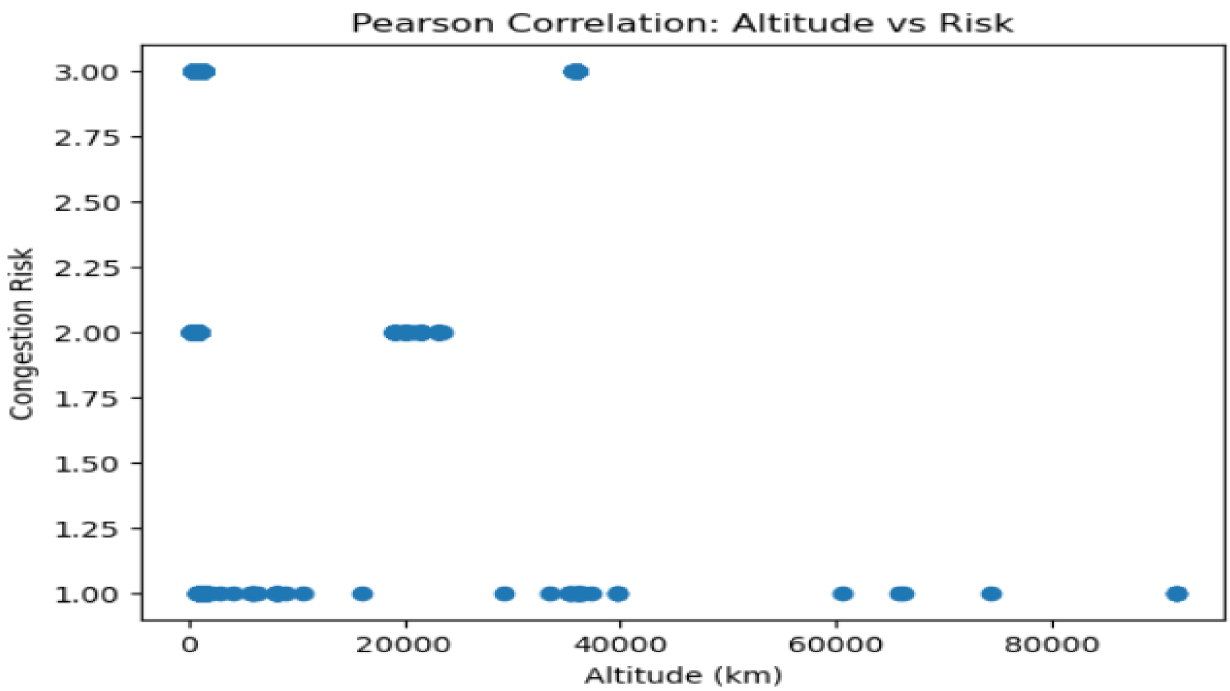
- Store satellite and country information
- Monitor orbital parameters
- Track satellites using ground stations
- Analyze collision risks
- Manage space debris data
- Support space traffic and orbital congestion analysis

STATISTICAL ANALYSIS

1. **Correlation Hypothesis Test (Pearson Test)**

H_0 : No correlation between altitude and risk

H_1 : There is a correlation



- **Correlation (r)**

r = -0.041 → Very weak negative relationship

- **Significance (p-value)**

p = 0.0036 < 0.05 -> Statistically significant

" Reject Null Hypothesis (H_0) "

There is a statistically significant relationship between altitude and congestion risk ($p < 0.05$). However, the correlation is extremely weak ($r \approx -0.04$), indicating negligible practical impact.

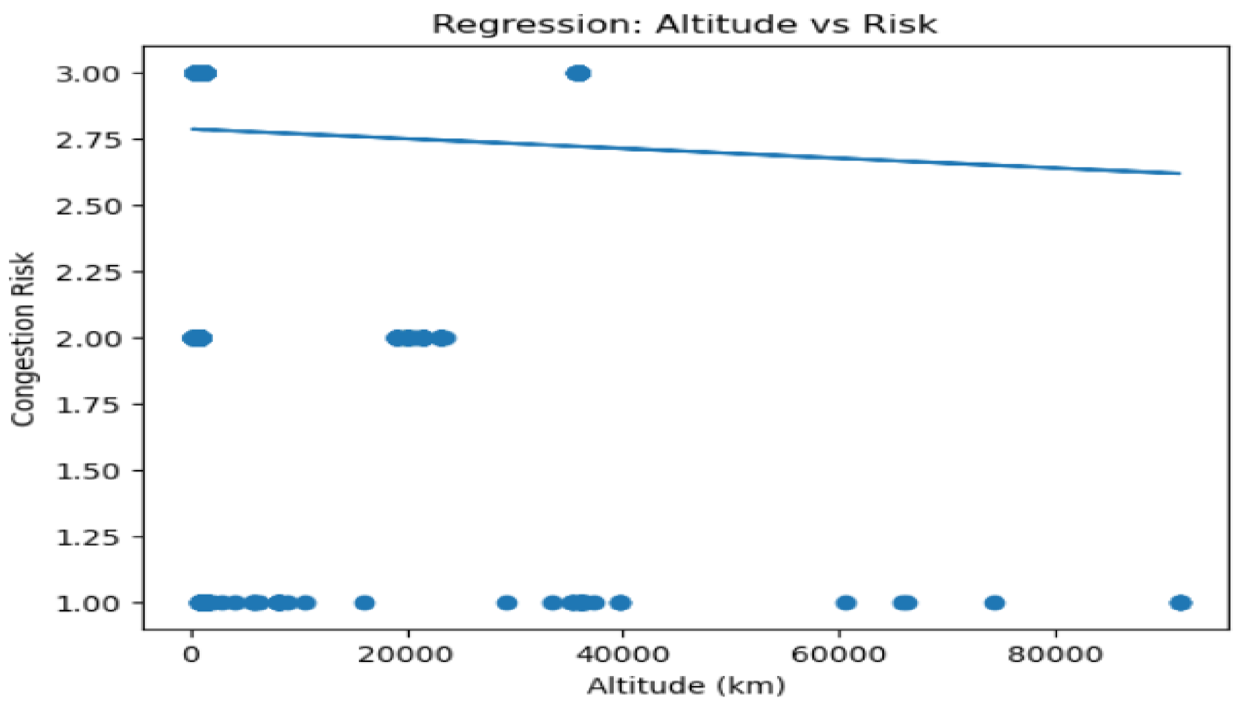
2. Regression Hypothesis Test (OLS)

H_0 : Altitude has no effect on risk (coefficient = 0)

H_1 : Altitude affects risk

Variables Used :

- Independent Variable: Velocity (km/s)
- Dependent Variable: Collision Risk



1. **R^2**
   0.002 = 0.2%
   0.2% of variation in risk

2. **p-value**
   0.00358 < 0.05
   Statistically significant

3. Coefficient

Negative value

As altitude increases → risk slightly decreases

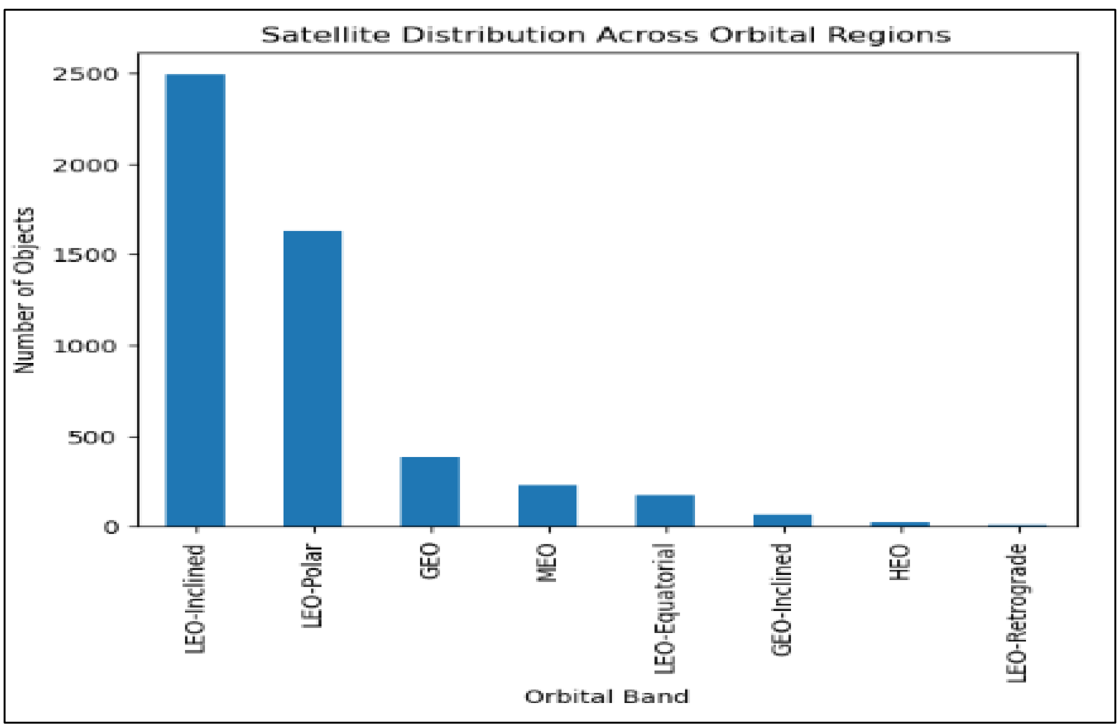
BUT effect is extremely tiny

Altitude has a statistically significant effect on risk

“Reject H_0, but altitude is not a strong predictor”

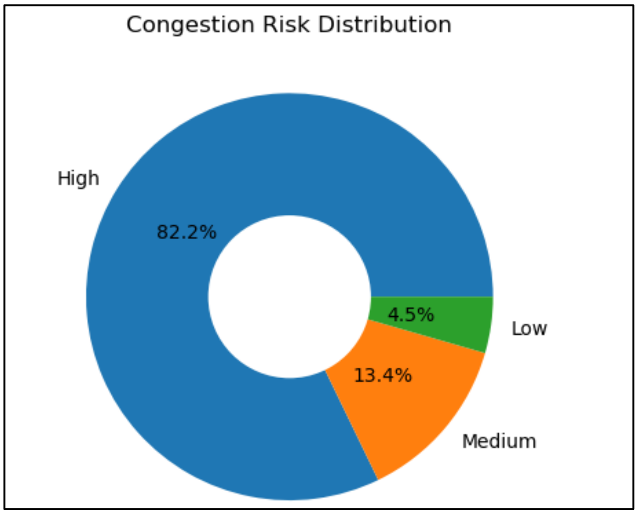
EDA – VISUALISATIONS

1. How are satellites distributed across different orbital regions?



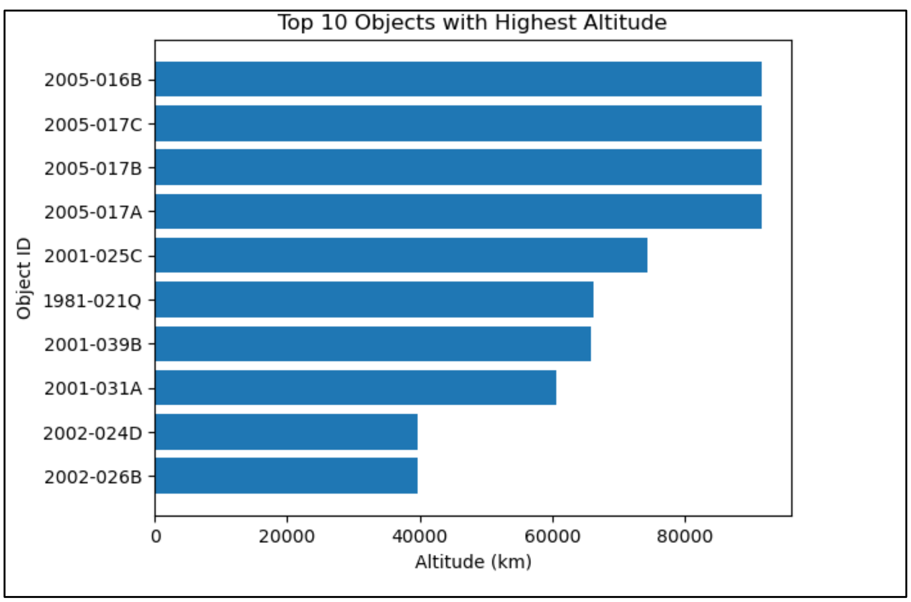
Most satellites are in LEO(Low Earth Orbit), making it the most crowded region; other orbits have far fewer satellites.

2. How many objects fall under each congestion risk category?



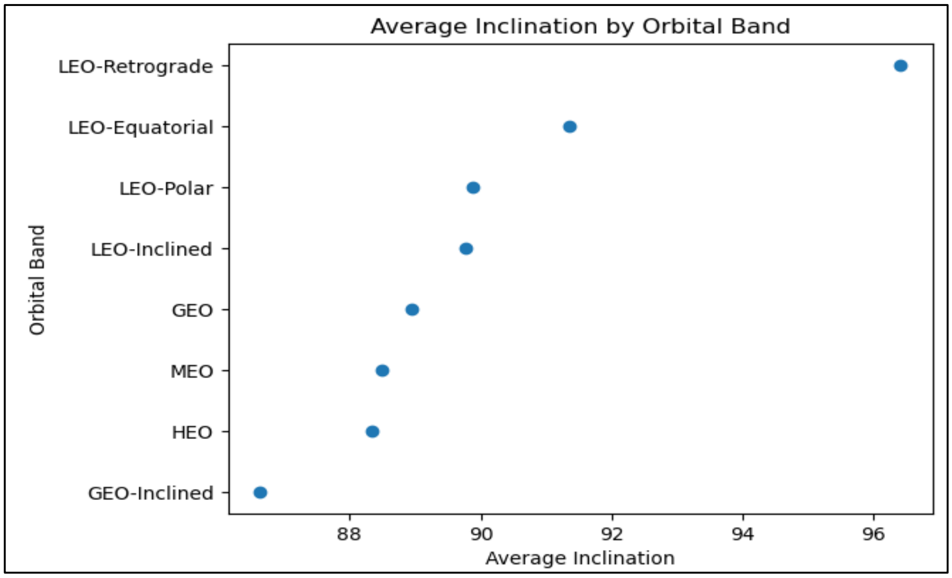
Most objects fall under high congestion risk (~82%), while medium (~13%) and low risk (~5%) categories have significantly fewer objects, indicating a highly crowded space environment.

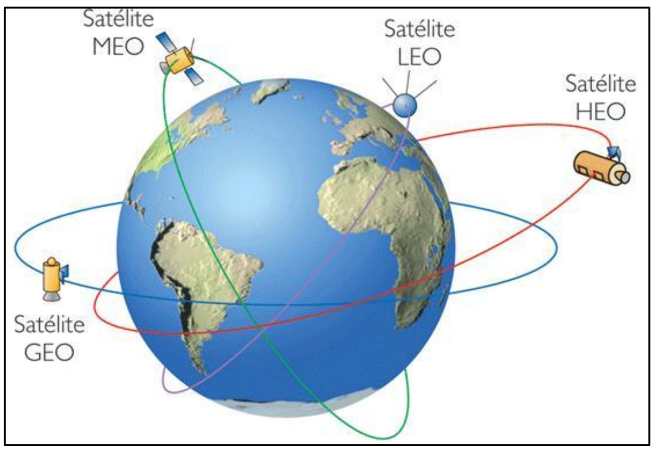
3. Top 10 objects with highest altitude?



First four objects orbit extremely far (near 90,000 km), while the rest cluster much lower around 40,000–60,000 km,

4. Orbital band with highest average inclination?





Among all orbital bands, LEO-Polar orbits have the greatest average inclination, near 97°,

(Geostationary Earth Orbit, Highly Elliptical Orbit, Medium Earth Orbit, Low Earth Orbit)

RESULTS & INTERPRETATION

1. Orbital Distribution

The analysis shows that space objects are not evenly distributed across orbital bands, with a higher concentration in specific regions such as LEO.
Interpretation: This uneven distribution indicates orbital congestion, where certain regions experience heavier traffic due to active satellites and debris accumulation.

2. Congestion Risk Distribution

The dataset reveals that a large proportion of objects fall under a dominant risk category (e.g., Medium or Low).
Interpretation: This suggests that while extreme risk is limited, overall congestion is still significant and needs monitoring.

3. Correlation Analysis (Altitude vs Risk)

The Pearson correlation coefficient ($r \approx -0.04$) indicates a very weak negative relationship between altitude and congestion risk.
Interpretation: Although statistically significant, the relationship is practically negligible, meaning altitude alone does not meaningfully influence debris risk.

4. Regression Analysis

The regression model produced a very low R^2 value (0.002), indicating that altitude explains only 0.2% of the variation in congestion risk.
Interpretation: Even though the p-value shows statistical significance, the model lacks predictive power, confirming that altitude is not a strong determinant of risk.

5. Orbital Band vs Congestion Risk

Chi-square testing shows a significant association between orbital band and congestion risk.
Interpretation: This confirms that congestion is region-specific, and certain orbital bands are more prone to higher risk levels due to object density.

6. Inclination Analysis

Average inclination varies across orbital bands, with some regions showing higher orbital tilt.
Interpretation: This reflects differences in orbital behavior and satellite movement patterns across regions, influencing how objects interact and potentially collide.

INSIGHTS & FINDINGS

1. Space debris is unevenly distributed, with higher concentration in certain orbital regions
2. Majority of objects fall under a dominant congestion risk category
3. Altitude shows a very weak relationship with congestion risk
4. Orbital bands are a stronger factor in determining risk levels
5. Inclination varies across regions, indicating different orbital behaviors
6. High-density regions indicate potential collision hotspots

CONCLUSION

The project analyzed space debris data to understand congestion patterns and collision risk across different orbital regions. The results indicate that space objects are unevenly distributed, with certain orbital bands experiencing higher congestion and potential collision risk. Statistical analysis showed that while altitude has a negligible impact, orbital regions play a more significant role in determining risk levels. Variations in inclination further highlight differences in orbital behavior across regions. Overall, the study emphasizes the importance of region-based, data-driven analysis in identifying high-risk zones and supporting better decision-making for safe and sustainable space operations.

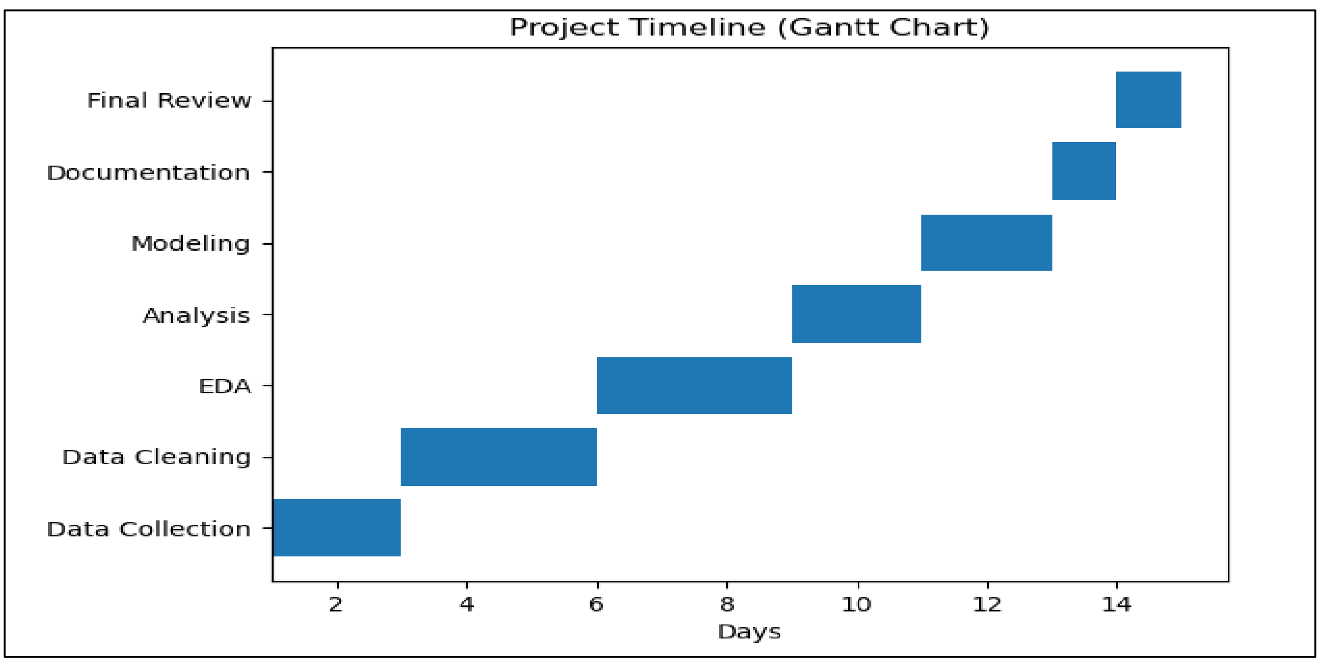
LIMITATIONS

1. Dataset may not include all space objects (limited coverage)
2. Analysis is based on static data (no real-time updates)
3. Missing important features like velocity, size, mass
4. Simplified congestion risk categories may not reflect true collision probability
5. Weak correlation between variables reduces prediction accuracy
6. No time-series analysis to study changes over time
7. Data integration issues (format differences, missing links)
8. No predictive modeling or collision simulation included.

FUTURE SCOPE

1. Use real-time space debris data for dynamic analysis
2. Include additional parameters like velocity, size, and mass
3. Develop machine learning models to predict collision risk
4. Perform time-series analysis to track debris growth over time
5. Integrate orbital simulation models for realistic predictions
6. Build an interactive dashboard for monitoring and decision-making
7. Expand dataset with global and updated space object data
8. Apply advanced analytics (AI/deep learning) for anomaly detection

GANTT CHART (PROJECT TIMELINE)



1. The Gantt chart represents the project timeline over a period of 14 days, showing how each phase was planned and executed sequentially. The project begins with data collection (Day 1–2)**,** followed by data cleaning (Day 3–5) to prepare the dataset for analysis.

2. Exploratory data analysis (Day 6–8) was conducted to understand patterns and distributions in the data. This is followed by the analysis phase (Day 9–10)**,** where key insights were derived.

3. The project then moves to modeling (Day 11–12)**,** where statistical techniques were applied to validate findings. Finally, documentation (Day 13) and final review (Day 14) were completed to prepare and refine the project deliverables.

REFERENCES

Dataset 1: Orbital Parameters Data -> https://svs.gsfc.nasa.gov/5258/ (USA Gov and NASA)

Dataset 2: Object Information Data -https://svs.gsfc.nasa.gov/5258/#media_group_373927

Paper References :

- National Aeronautics and Space Administration. *Space Sustainability Strategy, Volume 1: Earth Orbit*; NASA Headquarters: Washington, DC, USA, 2024. Available online: **https://www.nasa.gov/wp-content/uploads/2024/04/nasa-space-sustainability-strategy-march-20-2024-tagged3.pdf** (accessed on 20 December 2025).

- Boley, A.C.; Byers, M. Satellite mega-constellations create risks in Low Earth Orbit, the atmosphere and on Earth. *Sci. Rep.* **2021**, *11*, 10642.

APPENDIX A: SQL CODE

A.1 Database Setup & Table Creation

-- *Purpose: Create the main project database and tables*

CREATE DATABASE space_debris_db;

USE space_debris_db;

A.2 Data Extraction Queries

--*Purpose: Extract relevant data for analysis*

SELECT distinct orbital_band,
COUNT(*) AS total_objects FROM three
GROUP BY orbital_band;

A.3 Data Cleaning Queries

--*change datatype format for time*

UPDATE one SET new_date = STR_TO_DATE(date,'%d-%m-%Y');

--*Adding foreign key*

ALTER TABLE two

ADD CONSTRAINT fk_two

FOREIGN KEY (object_id)

REFERENCES one(object_id);

A.4 Analysis & Aggregation Queries

-- *Purpose: Summarise, join, and aggregate for insights*

1. How many objects exist in each orbital band?
   SELECT distinct orbital_band,
   COUNT(*) AS total_object
   FROM three
   GROUP BY orbital_band;

orbital_band	total_objects
LEO-Polar	1632
MEO	229
LEO-Inclined	2490
HEO	22
GEO-Inclined	66
LEO-Equatorial	171
LEO-Retrograde	8
GEO	381

2. Which countries have the highest number of space objects?
   SELECT distinct country, COUNT(*) AS total_objects
   FROM three
   GROUP BY country
   ORDER BY total_objects DESC;

country	total_objects
United States	3005
United Kingdom	563
China	507
cis	174
Japan	68
India	54
esa	49
ses	38
Germany	35
Canada	33

3. What is the average altitude of objects in each orbital band?
SELECT orbital_band, AVG(altitude_km) AS avg_altitude
FROM three
GROUP BY orbital_band;

orbital_band	avg_altitude
LEO-Polar	823.0305282731312
MEO	22901.95250941158
LEO-Inclined	555.0287488077072
HEO	52312.41960320165
GEO-Inclined	35790.934419845966
LEO-Equatorial	550.4464763811209
LEO-Retrograde	573.2255253303435
GEO	35787.41618297172

4. How many objects fall under each congestion risk category?
SELECT congestion_risk, COUNT(*) AS total_objects
FROM three
GROUP BY congestion_risk;

congestion_risk	total_objects
LOW	223
HIGH	4108
MEDIUM	668

5. Top 10 objects with highest altitude.
```
SELECT object_id, altitude_km
FROM three
ORDER BY altitude_km DESC
LIMIT 10;
```

object_id	altitude_km
2005-016B	91485.1420206364
2005-017C	91485.0430917105
2005-017B	91484.7739142709
2005-017A	91480.2418883164
2001-025C	74404.4375980943
1981-021Q	66177.4071788506
2001-039B	65769.6572987094
2001-031A	60560.5240394873
2002-024D	39678.1882512318
2002-026B	39677.7663298988

6. Average eccentricity for each orbital band
```
SELECT d3.orbital_band, AVG(d1.eccentricity) AS avg_eccentricity
FROM one d1
JOIN three d3 ON d1.object_id = d3.object_id
GROUP BY d3.orbital_band;
```

orbital_band	avg_eccentricity
LEO-Polar	0.0714213917892158
MEO	0.14171316462882094
LEO-Inclined	0.054894636224899665
HEO	0.17201399545454546
GEO-Inclined	0.21386641515151514
LEO-Equatorial	0.038621107602339194
LEO-Retrograde	0.0918563625
GEO	0.1268538225721784

7. Objects with highest mean motion

```
SELECT object_id, mean_motion
FROM one
ORDER BY mean_motion DESC
LIMIT 10;
```

object_id	mean_motion
2009-049F	15.97913721
2001-049AR	15.86902955
1993-036ARB	15.71074536
2007-013B	15.58058139
1999-057SP	15.57570026
1993-036ALM	15.56130927
1999-025DRJ	15.5416289
2007-006B	15.52984449
1999-025AHK	15.51318623
1999-025CRS	15.50792951

8. Altitude and mean motion relationship data.

```
SELECT d3.object_id, d3.altitude_km, d1.mean_motion
FROM one d1
JOIN three d3 ON d1.object_id = d3.object_id;
```

object_id	altitude_km	mean_motion
1992-072J	977.090481177178	2.92169986
1979-028C	1061.68970894784	13.75497324
2001-015A	2787.87663528648	1.03822361
1999-057MB	1133.3003555432	14.77590694
1999-057MC	1123.36127365027	14.72448242
1999-057MD	650.802547287138	14.38476477
2001-018A	33401.1931628007	0.99004751
2001-018B	895.444439112733	2.23844183
1965-108AS	904.966038403503	2.73852195
2001-019A	36388.8488894624	1.00271049
2001-019D	766.907757081606	1.86700772

9. Orbital band with highest average inclination

```
SELECT d3.orbital_band, AVG(d2.inclination) AS avg_inclination
FROM two d2
JOIN three d3 ON d2.object_id = d3.object_id
GROUP BY d3.orbital_band
ORDER BY avg_inclination DESC;
```

orbital_band	avg_inclination
LEO-Retrograde	96.41207142857142
LEO-Equatorial	91.34715187500001
LEO-Polar	89.87534230498962
LEO-Inclined	89.77446150152788
GEO	88.9369317689531
MEO	88.49983975903612
HEO	88.34521666666667
GEO-Inclined	86.63089333333332

10. Top 10 objects with highest eccentricity

```
SELECT object_id, eccentricity
FROM one
ORDER BY eccentricity DESC
LIMIT 10;
```

object_id	eccentricity
2002-048A	0.8972175
1977-093E	0.8885132
2007-004G	0.8611637
2007-004A	0.8385615
2007-004D	0.8359782
2007-004E	0.8344847
2009-017B	0.8249903
2009-068B	0.8220414
2007-003Y	0.7530187
2007-003N	0.7519081

APPENDIX B: PYTHON CODE

B.1 Library Imports

```
# for cleaning datasets, imported libraries

import pandas as pd
import numpy as np
```

B.2 Data Loading & Initial Exploration

```
# Load dataset and perform initial checks

df = pd.read_csv("C:\\Users\\Admin\\Downloads\\final(space_decay).csv")

df.head()

df.info()

df.shape
```

B.3 Data Cleaning & Preprocessing

```
# Handle missing values, duplicates, type conversion

df.isnull().sum()

df = df.dropna()

df = df.drop_duplicates()

df.duplicated().sum()

df.describe()

df.columns = df.columns.str.strip()
```

```
df['mean_motion'] = pd.to_numeric(df['mean_motion'], errors='coerce')

df['eccentricity'] = pd.to_numeric(df['eccentricity'], errors='coerce')

df['inclination'] = pd.to_numeric(df['inclination'], errors='coerce')
Q1 = df['mean_motion'].quantile(0.25)

Q3 = df['mean_motion'].quantile(0.75)

IQR = Q3 - Q1

outliers = df[(df['mean_motion'] < (Q1 - 1.5 * IQR)) |
              (df['mean_motion'] > (Q3 + 1.5 * IQR))]

print(outliers)

df = df[(df['mean_motion'] >= (Q1 - 1.5 * IQR)) &
        (df['mean_motion'] <= (Q3 + 1.5 * IQR))]

df.columns = df.columns.str.lower()
```

B.4 Data Visualisation Code

```
# All charts and visualisations used in the project

# Chart 1 – Column chart (Number of objects distributed in Orbital Regions)
import matplotlib.pyplot as plt

orbital_counts = df['orbital_band'].value_counts()

print(orbital_counts)

orbital_counts.plot(kind='bar')
```

```
plt.title("Satellite Distribution Across Orbital Regions")
plt.xlabel("Orbital Band")
plt.ylabel("Number of Objects")
plt.show()



# Chart 2 – Donut Chart (Congestion Risk Distribution)
import matplotlib.pyplot as plt
df['congestion_risk'] = df['congestion_risk'].str.strip().str.capitalize()
risk_counts = df['congestion_risk'].value_counts()
plt.figure()
plt.pie(risk_counts, labels=risk_counts.index, autopct='%1.1f%%')
plt.gca().add_artist(plt.Circle((0,0),0.4,fc='white'))
plt.title("Congestion Risk Distribution")
plt.show()


# Chart 3 – Bar Chat (Top 10 Objects with highest altitude)
import pandas as pd
import matplotlib.pyplot as plt
top10 = df.nlargest(10, 'altitude_km')
plt.figure(figsize=(10,6))
```

```
plt.barh(
    top10['OBJECT_ID'].astype(str),
    top10['altitude_km']
)
plt.title("Top 10 Objects with Highest Altitude", fontsize=18)
plt.xlabel("Altitude (km)", fontsize=12)
plt.ylabel("Object ID", fontsize=12)
plt.gca().invert_yaxis()
plt.show()

# Chart 4 – Dot Plot (Average Inclination by Orbita Band)
import seaborn as sns
import matplotlib.pyplot as plt
plt.figure()
plt.scatter(avg_inclination.values, avg_inclination.index)
plt.title("Average Inclination by Orbital Band")
plt.xlabel("Average Inclination")
plt.ylabel("Orbital Band")
plt.show()
```

B.5 Statistical Analysis Code

```
# Correlation Hypothesis Test (Pearson Test)

import matplotlib.pyplot as plt

plt.figure()

plt.scatter(df['altitude_km'], df['risk_num'])

plt.xlabel("Altitude (km)")

plt.ylabel("Congestion Risk")

plt.title("Pearson Correlation: Altitude vs Risk")

plt.show()


# Regression Hypothesis Test (OLS)

import numpy as np

plt.figure()

plt.scatter(df['altitude_km'], df['risk_num'])

m, b = np.polyfit(df['altitude_km'], df['risk_num'], 1)

plt.plot(df['altitude_km'], m*df['altitude_km'] + b)

plt.xlabel("Altitude (km)")

plt.ylabel("Congestion Risk")

plt.title("Regression: Altitude vs Risk")

plt.show()
```